

Research Methodology in CS (CSEG3060_3)

Lecture 1: Understanding Research Problem: Meaning and Significance, Sources and Identification

Unit 1: Introduction to Research Problem • Mapped Course Outcome: CO1

Lecture 1: Understanding Research Problem — Meaning, Significance, Sources, and Identification

Course: Research Methodology in CS (CSEG3060_3) **Program:** B.Tech (Computer Science)
Semester: 5 **Unit:** 1 — Introduction to Research Problem **Mapped Course Outcome:** CO1 **Lecture Duration:** 60 minutes (recommended two-slot delivery)

1. Lecture Overview

Research, in its most elemental sense, is the systematic and rigorous pursuit of knowledge through structured inquiry. In computer science, where the pace of technological evolution is rapid and the boundary between applied and theoretical work is increasingly porous, the formulation of a clear and meaningful research problem is arguably the single most consequential decision a researcher makes. A poorly conceived problem statement can compromise the validity, feasibility, and contribution of an entire research effort, regardless of how sophisticated the subsequent methodology may be.

This opening lecture of Unit 1 establishes the conceptual foundation required to understand what constitutes a research problem, why its formulation carries significant weight, the diverse sources from which research problems emerge, and the systematic processes through which they can be identified and refined. The content directly underpins Course Outcome CO1, which concerns the student's ability to conceptualize, articulate, and justify a research problem in a computer science context.

1.1 Learning Objectives

By the end of this lecture, the student should be able to:

1. Define and distinguish a "problem" from a "research problem."
2. Articulate the academic, practical, and societal significance of a well-formulated research problem.

3. Enumerate and critically evaluate the primary sources of research problems in computer science.
4. Apply structured techniques for the identification and initial screening of research problems.
5. Recognize the typical errors committed during research problem selection.

1.2 Reading and References

- Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches* (5th ed.). SAGE.
 - Kothari, C. R. (2004). *Research Methodology: Methods and Techniques* (2nd ed.). New Age International.
 - Kumar, R. (2014). *Research Methodology: A Step-by-Step Guide for Beginners* (4th ed.). SAGE.
 - Sekaran, U., & Bougie, R. (2016). *Research Methods for Business* (7th ed.). Wiley.
 - Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in Software Engineering*. Springer.
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2. Key Concepts and Definitions

Before delving into the substance of the lecture, it is essential to anchor the discussion in precise definitions.

2.1 Problem

A **problem** is any situation, condition, or phenomenon that exhibits a discernible gap between an existing state (*what is*) and a desired state (*what ought to be*). In everyday language, a problem may be trivial, ambiguous, or merely descriptive.

2.2 Research Problem

A **research problem** is a specific, well-articulated, and researchable issue that arises from a recognized gap in knowledge, theory, or practice. It must be amenable to systematic investigation using scientific methods, and its investigation must contribute to the body of existing knowledge.

Formally, a research problem can be expressed as:

$$R = f(G, Q, F)$$

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where R denotes the research problem, G represents the identified gap or deficiency, Q encapsulates the research questions to be addressed, and F refers to the feasibility constraints

within which the inquiry is to be conducted.

2.3 Problem Statement

A **problem statement** is the textual and logical articulation of the research problem. It situates the problem within its disciplinary context, justifies its significance, and previews the general approach that will be adopted.

2.4 Research Question

A **research question** is a focused interrogative that operationalizes the problem statement by translating it into specific, answerable inquiries. Research questions typically address *what*, *how*, *why*, or *to what extent* dimensions of the phenomenon under study.

2.5 Distinguishing Research Problems from Research Topics

Aspect	Research Topic	Research Problem
Scope	Broad area of interest	Specific issue within the topic
Nature	Descriptive label	Investigative question
Formulation	Often a noun phrase	Usually an interrogative or declarative proposition
Investigation	Not directly investigable	Directly investigable using scientific methods
Example	Machine Learning in Healthcare	Mitigating demographic bias in chest X-ray classification models

This distinction is critical because students frequently begin their theses with topics rather than problems, leading to vague, unfocused, and ultimately unmanageable research designs.

3. Meaning of a Research Problem: A Detailed Analysis

3.1 The Ontological Position of a Research Problem

A research problem is not merely an inconvenience or a difficulty; it is a structured epistemic encounter with the unknown. In computer science, the unknown may manifest in many forms:

- A previously unstudied phenomenon (e.g., emergent behavior in large language model agents).
- An unresolved inconsistency in existing literature (e.g., conflicting performance claims of competing algorithms).

- A practical constraint that lacks a satisfactory solution (e.g., latency in real-time edge AI inference).
- A theoretical proposition awaiting empirical verification.

The research problem, therefore, occupies the ontological core of any research endeavor. It defines what is to be known and constrains how that knowing is to be pursued.

3.2 Components of a Well-Formulated Research Problem

A complete research problem formulation contains the following components:

1. **Problem Domain Identification:** The broad subject area within which the problem resides (e.g., natural language processing, distributed systems, software engineering).
2. **Gap Specification:** A precise articulation of what is missing, unresolved, or contradictory in current knowledge.
3. **Contextualization:** The situational, technological, or societal context that renders the problem relevant.
4. **Population or Subject Definition:** The entities (software systems, users, algorithms, datasets) that constitute the focus of investigation.
5. **Variable Identification:** The key constructs whose relationships or behaviors are to be examined.
6. **Preliminary Boundaries:** The scope constraints that delineate what is included and excluded.

3.3 The Hierarchical Structure of Inquiry

A useful mental model is the hierarchy that descends from broad theme to specific inquiry:

Research Theme → Research Topic → Research Problem → Research Question(s) → Research Obj



A scholar working on *deep learning for medical image analysis* (theme) may focus on *explainability of diagnostic models* (topic), identify *the lack of clinically interpretable explanations for transformer-based radiology models* (problem), translate this into *how do post-hoc explanation techniques compare in conveying clinically meaningful information to radiologists* (question), and operationalize it through *to evaluate three explanation techniques across 200 chest X-ray predictions* (objective).

This descending hierarchy is a powerful diagnostic tool for assessing the maturity of any research formulation.

4. Significance of the Research Problem

4.1 Why the Research Problem Deserves Singular Attention

The research problem is the fulcrum upon which the entire research project balances. The significance of devoting deliberate attention to its formulation can be enumerated across four dimensions.

4.1.1 Methodological Significance

The choice of problem dictates the choice of methodology. A problem seeking causal explanation demands experimental or quasi-experimental designs. A problem seeking in-depth understanding of user experience demands qualitative inquiry. A problem seeking measurable improvement in system performance demands benchmark-driven evaluation. An ill-defined problem often forces the researcher into inappropriate methodologies, producing results that are technically valid but practically irrelevant.

4.1.2 Theoretical Significance

A well-formulated research problem contributes to theory construction, refinement, or extension. In computer science, this may take the form of new algorithms, complexity bounds, formal models, or architectural frameworks. Without a strong problem foundation, the resulting theoretical contributions tend to be incremental, fragmented, or ad hoc.

4.1.3 Practical and Societal Significance

In an applied discipline such as computer science, research problems are often judged by their practical relevance. A problem that addresses an industry pain point (e.g., reducing false positives in fraud detection), a societal challenge (e.g., accessibility of digital interfaces for persons with disabilities), or an environmental concern (e.g., energy-efficient training of deep learning models) carries tangible impact.

4.1.4 Personal and Academic Significance

For the individual researcher, the research problem shapes the entire scholarly trajectory. A well-chosen problem sustains motivation through the inevitable setbacks of empirical work, opens pathways for publication, and forms the basis of doctoral or postdoctoral work.

4.2 The Cost of a Poorly Defined Problem

The literature in research methodology is replete with cautionary examples. The principal consequences of a poorly formulated research problem include:

- **Methodological mismatch:** Applying inappropriate instruments or techniques.
 - **Scope creep:** Continuous expansion that renders the project unfinishable.
 - **Invalid conclusions:** Drawing inferences that the data cannot support because the question was poorly framed.
 - **Wasted resources:** Months or years of effort on questions that the literature has already answered.
 - **Publication rejection:** Manuscripts are routinely rejected at the editorial stage for vagueness of problem.
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5. Sources of Research Problems

Research problems in computer science originate from a remarkably diverse set of sources. A mature researcher cultivates habits of attention across multiple channels.

5.1 Primary Sources

Primary sources are those directly encountered through the researcher's own engagement with the field.

5.1.1 Personal Experience and Practice

Practitioners who have spent years in software development, system administration, or data engineering often develop an intuitive sense of recurring inefficiencies, recurring defects, or unmet needs. For instance, a developer who has repeatedly encountered debugging challenges in microservices architectures may formulate a research problem on automated root cause analysis.

5.1.2 Direct Observation

Field observation of how users interact with technology, how developers collaborate, or how systems fail in production environments can surface research problems that are invisible to those who work only at the theoretical level.

5.1.3 Industrial Experience and Internships

Internships, capstone projects, and industry collaborations frequently expose students to practical constraints that have not been addressed in academic literature. These constraints are fertile ground for applied research problems.

5.2 Secondary Sources

Secondary sources are those accessed through the documented work of others.

5.2.1 Academic Literature

The most disciplined source of research problems. Published journal articles, conference proceedings, and review papers routinely identify gaps in the form of phrases such as "future work," "limitation," "unresolved issue," and "open question."

A systematic review of recent papers (e.g., the top-tier venues such as NeurIPS, ICSE, SIGCOMM, CCS, IEEE S&P) can reveal patterns of unresolved questions that aggregate into a coherent research problem.

5.2.2 Theses and Dissertations

Master's and doctoral theses often conclude with explicit suggestions for further research. These are valuable because they have already passed peer scrutiny.

5.2.3 Books and Monographs

Authoritative textbooks and research monographs articulate the foundational problems of a sub-discipline and trace their evolution. Foundational works in areas such as algorithm design (Cormen et al.), distributed systems (Tanenbaum and Van Steen), and software engineering (Sommerville) are particularly valuable.

5.2.4 Patents and Industrial Publications

Patents disclose novel technical solutions and, by implication, the problems they solve. Industrial white papers from organizations such as Google, Microsoft, IBM, and Meta often articulate research problems years before they appear in academic venues.

5.2.5 Funding Agency Calls and Research Agendas

Calls for proposals from funding bodies (e.g., DST, SERB, MEITY in India; NSF, DARPA, ERC internationally) articulate research priorities and explicitly identify problems of national or strategic importance. Researchers seeking funding alignment frequently derive their problems from such calls.

5.2.6 Standards Bodies and Regulatory Documents

Standards published by ISO, IEEE, NIST, and similar bodies often identify technical problems that require resolution to ensure interoperability, security, or performance.

5.3 Tertiary Sources

5.3.1 Discussions with Peers and Mentors

Supervisors, senior researchers, and fellow students often possess contextual knowledge that points to promising problems.

5.3.2 Conferences, Workshops, and Symposia

Live discussions at conferences frequently surface disagreements or unexplored intersections that constitute research problems. The questions raised during and after presentations are particularly valuable indicators of unresolved issues.

5.3.3 Online Communities

Forums such as Stack Overflow, ResearchGate discussions, arXiv comment threads, GitHub issue trackers, and specialized mailing lists often reveal recurring pain points and unmet needs.

5.4 Sources Particularly Relevant to Computer Science

Computer science enjoys a distinctive ecosystem of sources that merits explicit attention:

Source Type	Example	Typical Problem Type
Open-source issue trackers	GitHub issues on TensorFlow	Reproducibility, performance bugs
Benchmark leaderboards	Papers With Code	Performance gaps in specific tasks
Kaggle competitions	Past competition problems	Real-world data science challenges
Industry tech blogs	Google AI Blog, Meta Research	Emerging research directions
Government reports	AI safety reports	Sociotechnical problems
CVE databases	Common Vulnerabilities and Exposures	Cybersecurity research gaps

6. Identification of the Research Problem

The act of identifying a research problem is itself a methodical process. Identification differs from selection (which involves evaluation against criteria) in that it focuses on surfacing candidate problems from the broader landscape.

6.1 The Identification Process

A structured approach to identifying research problems may proceed through the following phases:

Phase 1: Broad Familiarization

The researcher immerses in the chosen domain through a broad survey of recent literature, textbooks, and conference proceedings. The objective is to develop a working mental map of the domain's contours.

Phase 2: Deliberate Gap Search

The researcher conducts targeted searches designed to surface gaps:

- Search for "future work," "limitation," "open challenge" in recent review papers.
- Examine the reference lists of seminal works to identify threads that have remained dormant.
- Compare competing approaches to identify unresolved trade-offs.

Phase 3: Question Articulation

For each candidate gap, the researcher articulates a provisional research question. A useful heuristic is to attempt to write the question in a single declarative sentence. If this proves impossible, the question is likely not yet mature.

Phase 4: Preliminary Feasibility Check

The researcher conducts a preliminary check on availability of data, instruments, expertise, and time. Problems that fail this check are set aside or deferred.

Phase 5: Refinement and Crystallization

The surviving candidate problems are refined through iterative writing, peer discussion, and supervisor consultation until a single, well-articulated problem emerges.

6.2 Techniques for Problem Identification

Several structured techniques assist in the identification phase:

6.2.1 Literature-Driven Identification

The most common technique. It involves:

- Conducting a **systematic literature review** or **scoping review**.
- Using tools such as **PRISMA** (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) to document the search and selection process.
- Synthesizing findings into a conceptual map that visualizes the structure of knowledge and its gaps.

6.2.2 Curiosity-Driven Identification

A less structured but often more creative technique. The researcher maintains a research journal in which observations, questions, and intuitions are recorded over weeks or months. Patterns that emerge from this journal frequently crystallize into research problems.

6.2.3 Delphi Technique

A structured communication method in which a panel of experts answers questionnaires in multiple rounds. The anonymity of the process reduces conformity pressure and often surfaces non-obvious problems. Particularly useful for identifying research problems at the intersection of disciplines.

6.2.4 Brainstorming and Focus Groups

Used less frequently in computer science but valuable for identifying research problems in human-centered computing, HCI, and educational technology. Groups of practitioners, users, or researchers generate candidate problems, which are then clustered and prioritized.

6.2.5 Problem Tree Analysis

A technique borrowed from development research. The researcher identifies the focal issue, then enumerates its causes (roots) and effects (branches). The research problem may emerge from the prioritization of one of these causes or effects.

6.2.6 Concept Mapping

Visual representations of relationships among concepts within a domain. Concept maps frequently reveal clusters of densely interconnected concepts separated from other clusters by sparse connections. The sparse connections often indicate research opportunities.

6.3 A Note on Artifactual vs. Natural Problems

In software engineering research, Wohlin et al. (2012) distinguish between:

- **Natural problems:** Problems encountered in the wild, typically through industry collaboration.
- **Artifactual problems:** Problems constructed for the purpose of experimentation, typically through controlled settings.

Each type has implications for the external validity of the resulting research. Researchers should consciously identify which type they are pursuing and structure their methodology accordingly.

7. Illustrative Case Studies in Computer Science

The following case studies illustrate how research problems in computer science emerge from real sources and are refined into actionable inquiries.

7.1 Case Study A: Explainability in Deep Learning Models for Medical Diagnosis

Source: Published literature on deep learning for radiology, supplemented by clinical observation during a hospital internship.

Original Gap Identified: Deep learning models achieve high diagnostic accuracy but provide outputs that radiologists find difficult to interpret, limiting clinical adoption.

Refined Research Problem: "How do post-hoc explanation methods (Grad-CAM, LIME, SHAP) compare in supporting diagnostic reasoning by radiologists using transformer-based chest X-ray classifiers?"

Significance: Addresses the practical gap between algorithmic performance and clinical usability.

7.2 Case Study B: Energy Efficiency in Large Language Model Training

Source: Industry white papers and personal observation during a research internship at a major AI laboratory.

Original Gap Identified: Training of large language models consumes substantial energy, but the precise contribution of various architectural choices to overall energy consumption is poorly

characterized.

Refined Research Problem: "What is the relative contribution of attention mechanism variants to the energy consumption of transformer training, and how does this interact with model size?"

Significance: Addresses the intersection of algorithmic design and environmental sustainability.

7.3 Case Study C: Technical Debt in Machine Learning Systems

Source: Industry experience combined with surveys of software engineering literature.

Original Gap Identified: Traditional technical debt frameworks do not adequately capture ML-specific debts (data dependencies, model staleness, feedback loops).

Refined Research Problem: "How can existing technical debt taxonomies be extended to systematically capture ML-specific debt in production systems, and what evidence supports the prevalence of each category?"

Significance: Bridges software engineering and AI/ML practice, with direct industrial relevance.

8. Common Errors in Research Problem Selection (Preview)

Although a detailed treatment of errors belongs to the next lecture, a preliminary awareness of common pitfalls is valuable:

1. Selecting a problem that is **too broad** to be investigated within the available resources.
2. Selecting a problem that has **already been conclusively addressed** in the literature.
3. Selecting a problem driven primarily by **technical fashion** rather than genuine intellectual or practical need.
4. Selecting a problem whose **data or instruments are inaccessible**.
5. Selecting a problem that is **insignificant** — interesting to the researcher but immaterial to the field.
6. Selecting a problem that is **infeasible** due to ethical, technical, or logistical constraints.

These errors will be revisited in greater depth in the subsequent lecture.

9. Summary

This lecture established the foundational concepts necessary for the disciplined formulation of research problems in computer science. The key takeaways are:

- A **research problem** is a specific, researchable articulation of a gap between existing and desired knowledge. It is more precise and methodologically tractable than a general topic.
- The **significance** of a research problem is multidimensional: methodological, theoretical, practical, and personal. A well-formulated problem aligns the researcher with meaningful contribution.
- **Sources of research problems** are diverse, spanning personal experience, observation, academic literature, industrial practice, funding calls, and online communities. Computer science enjoys a particularly rich source ecosystem through open-source repositories, benchmark platforms, and tech industry publications.
- **Identification** is a structured process involving broad familiarization, gap search, question articulation, feasibility assessment, and iterative refinement. Techniques such as systematic literature review, the Delphi method, and concept mapping assist in this process.
- A **clear hierarchy** descends from research theme to topic to problem to question to objective, providing a useful diagnostic for research formulation maturity.

The next lecture will examine the criteria and characteristics of a good research problem, the essential attributes that distinguish meritorious problems, and the typical errors encountered in the selection process.

10. Exam Preparation and Practice

10.1 Key Terms for Short-Answer Questions

Students should be able to define and differentiate the following terms without reference:

- Problem vs. research problem
- Research problem vs. research question
- Problem statement vs. problem formulation
- Primary, secondary, and tertiary sources
- Systematic literature review
- PRISMA
- Delphi technique
- Concept mapping
- Natural vs. artifactual problems

10.2 Sample Short-Answer Questions (2–5 marks each)

1. Differentiate between a research topic and a research problem. Illustrate with a computer science example.
2. Why is the formulation of a research problem considered the most critical step in the research process?
3. Enumerate four sources of research problems relevant to computer science and briefly explain each.
4. Explain the role of literature review in identifying a research problem.
5. What is the Delphi technique, and how does it assist in problem identification?
6. Distinguish between natural and artifactual problems in software engineering research.

10.3 Sample Long-Answer Questions (10–15 marks)

1. Discuss the meaning and significance of a research problem in computer science. With suitable examples, illustrate how a broad research area is progressively narrowed into a well-defined research problem.
2. Critically examine the various sources from which research problems in computer science may be identified. Which sources do you consider most reliable, and why?
3. Describe the structured process of identifying a research problem. Explain at least three techniques that may be employed during the identification phase, with examples from computer science domains.
4. A student wishes to undertake research on "artificial intelligence in healthcare." As a research methodology expert, guide the student through the formulation of a well-defined research problem, citing appropriate sources and techniques.

10.4 Applied Exercise

Select any one of the following broad themes from computer science:

- Generative AI for software engineering
- Blockchain-based identity management
- Quantum-resistant cryptographic algorithms
- Edge AI for autonomous systems
- Privacy-preserving machine learning

Within one week, deliver a written submission (maximum 1,500 words) that:

1. Identifies at least three candidate research problems within the chosen theme, citing the sources from which each was identified.
2. Refines one of these problems into a single, well-articulated research problem statement.
3. Justifies the significance of the chosen problem across academic, practical, and societal dimensions.
4. Provides a preliminary assessment of feasibility with respect to data, instrumentation, and time.

10.5 Reflective Prompt

Before the next lecture, maintain a research journal in which you record, over the course of one week, every instance in which you encounter a difficulty, limitation, or unresolved question in your daily academic or computational work. Bring this journal to the next session, where it will serve as the basis for the first classroom exercise in problem identification.

End of Lecture 1 Notes — Unit 1: Introduction to Research Problem.